CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B.Tech Examination (IT)

May 2015

IT214 Data Structures and Files

Date: 06.05.2015,Wednesday **Time: 01:30 pm to 04:30 pm** **Maximum Marks:70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1 Choose Most Appropriate Answer from the following.

(1) A graph in which there exists an edge between every pair of nodes is known as…..... **1**

(A) Complete Graph (B) Planner Graph

(C) Regular Graph (D) Non Regular Graph

(2) Maximum number of edges in an undirected graph with n nodes without self-loops is__________. **1**

(A) n^2 (B) n (n-1) / 2

(C) $(3n^2-3n)/2$ (D) none of the above

(3) In linked representation of Binary trees LEFT [k] contains the........ of at the node N, where k is the location. **1**

(A) Data (B) Location and left child

(C) Right child address (D) Null Value

(4) What is the minimum number of nodes in a full binary tree with depth 3? **1**

(A) 3 (B) 8 (C) 11 (D) 15

(5) **1**

(b) What is linear and nonlinear data structure? Explain with suitable example. **2**

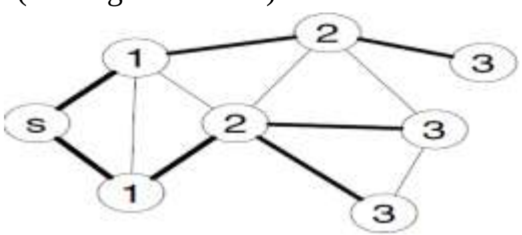
Q-2

(a) Define the Following terminology: **4**

Multi Graph, Degree of Vertices, Complete Binary Tree, AVL Tree.

(b) **Answer following questions.(Any Two)** **10**

1 Apply DFS on below graph. (starting vertex is S)



2 Arrange the following data in the array using hash function H(x). Calculate hash key and use linear probing to resolve collisions. Array size is 8 (index starts from 0)

Data: 214, 4563, 789, 2477, 1579, 4566, 1000, 6745

Hash function H(x) = (x modulo 5) + 1

3 Construct the Binary Tree from below Preorder and In order Sequence.
Pre-order: **23, 18, 12, 20, 19, 44, 35, 38, 52.**
In-order**: 12, 18, 19, 20, 23, 35, 38, 44, 52.**
Write Post order Sequence from the generated Binary Tree.

Q-3 Attempt Any Two. **14**

(a) Create Binary Search Tree of below inputs.
Input Sequence = **40, 20, 30, 60, 70, 10, 50, 55, 35, 15, 75, 100, 110, 200, 150.**
After creation, apply deletion operation on node 40, 50, 75, 200 and 100. Show the BST after each Deletion.
If Each deletion takes 2 nanoseconds then, calculate the total time of five deletion operations.

(b)

(c) Construct an AVL Tree from given inputs.
Input Sequence : **21, 26, 30, 9, 4, 14, 28, 18, 15, 10, 2, 3, 7.**
Each insertion costs 3 machine cycle and adjustment of an element costs 5 machine cycle. Calculate the total machine cycles needed to construct an AVL Tree.

SECTION-II

Q-4 Choose Most Appropriate Answer from the following.

(a)

1 Suppose a circular queue of capacity (n – 1) elements is implemented with an array of n elements. Assume that the insertion and deletion operation are carried out using REAR and FRONT as array index variables, respectively. Initially, REAR = FRONT = 0. The conditions to detect queue full and queue empty are **1**

(a) Full: (REAR+1) mod n == FRONT, empty: REAR == FRONT
(b) Full: (REAR+1) mod n == FRONT, empty: (FRONT+1) mod n == REAR
(c) Full: REAR == FRONT, empty: (REAR+1) mod n == FRONT
(d) Full: (FRONT+1) mod n == REAR, empty: REAR == FRONT

2 Which of the following is an application of stack? **1**

(a) Finding factorial
(b) Tower of Hanoi
(c) Infix to postfix
(d) All of the above

3 If the sequence of operations – push(a), push(z), pop, push(a), push(z), pop, pop, pop, push(z), pop are performed on a stack, the sequence of popped out values are? **1**

(a) z, z, a, a, z
(b) z, z, a, z, z
(c) z, a, z, z, a
(d) z, a, z, z, z

4 The result evaluating the postfix expression 10 5 + 60 6 / * 8 – 8 is **1**

(a) 284
(b) 213
(c) 142
(d) 71

5 Following is C like pseudo code of a function that takes a Queue as an argument, and uses a **2**

stack S to do processing

```
void fun(Queue *Q)
{
   Stack S;  // Say it creates an empty stack S

   // Run while Q is not empty
   while (!isEmpty(Q))
   {
      // deQueue an item from Q and push the dequeued item to S
      push(&S, deQueue(Q));
   }

   // Run while Stack S is not empty
   while (!isEmpty(&S))
   {
     // Pop an item from S and enqueue the poppped item to Q
     enQueue(Q, pop(&S));
   }
}
```

What does the above function do in general?

(a) Removes the last from Q
(b) Keeps the Q same as it was before the call
(c) Makes Q empty
(d) Reverses the Q

6 Sort the following input using bubble sort, what will be the output string after 3rd pass.
Input Sequence = 11, 23,5,7,3,10,8.

(A) 5,7,3,10,8,11,23 (B) 5,3,7,8,10,11,23
(C) 3,5,7,8,10,11,23 (D) 3,5,8,7,10,11,23

7 In a selection sort of n elements, how many times is the swap function called in the complete execution of the algorithm?

(A) 1 (B) n-1 (C) n $\log_2 n$ (D) n^2

2

Q-5

(a) Given a two dimensional array A(-12 : 1, 3 : 8) stored in row-major order with base address of 3000 and size of each element is 4 bytes, Find the address of an element A(-5 , 6) **4**

(b) **Answer following questions.(Any Two)** **10**

1 (i) Convert the following infix expression into its equivalent postfix expression using stack. **3**

m + (n - (a ^ (b – c) * d) / f)

(ii) Evaluate the given postfix expression using stack. **2**

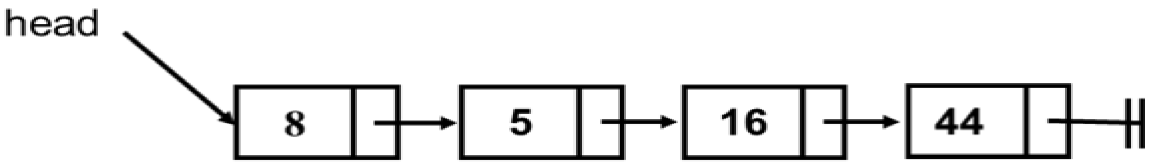
(2 + 3) * 4 + 5 / (3 + 2 * 4) + 5

2 Show the content of circular queue with front and rear pointer after each operation. Initially queue is empty. Size of queue is 5. The sequence of operation given below: **5**

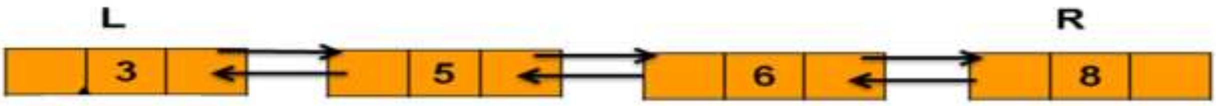
- Insert 10, 50, 40, 80
- Delete

- Insert 200, 70, 150
- Delete
- Delete
- Delete
- Delete

3	What is Tower of Hanoi Problem? Trace its recursive solution for disk= 3	**5**
Q-6	**Attempt Any Two**	**14**
(a)	Apply Insertion Sort on below string and find out the no. of comparison and no. of exchanges. Input Sequence = **2, 9, 7, 6, 15, 16, 5, 10, 11.**	
(b)	List the advantages of doubly linked list over singly linked list.	**3**
	Write the step to delete node x from given singly linked list. (Delete operation contains two parameter Delete (x, head) where x = 5).	**4**



(c)	Write the steps to insert a new node with info = 2 as left most element and a node with info = 9 as right most element to the given doubly linked list. (Where Leftmost node and rightmost node addresses are given by the pointer variables ***L*** and ***R*** respectively. ***NEW*** is the address of new node. The left and right links of a node are denoted by ***LPTR*** and ***RPTR*** respectively. Draw the position of each node along with their pointers after every step.)	



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